HDPT Bus System Strategy Proposal

MCC Group Presentation

Grady Allen, Madi Cohen, Zach Runkle, Elena Truitt

HDPT
hdpt.com

JMU

MADISON CONSULTING CLUB

Case Overview

About HDPT

The **Harrisonburg Department of Public Transportation (HDPT)** provides essential bus services for **JMU students, faculty, and staff**, supporting both **on-campus and off-campus** mobility. While buses are a **sustainable** and **cost-effective** transit option, operational inefficiencies impact overall service quality.

Problem Statement

The HDPT Bus System at JMU struggles with overcrowding, underutilized routes, long wait times, inconsistent scheduling, limited real-time tracking, and rising costs, **reducing reliability**, **increasing congestion**, and **hindering the rider experience**.

Strategic Approach

To improve HDPT's operational strategy, our proposal focuses on **data-driven optimization, enhanced rider experience, and cost-efficient innovations.**

Cost Reduction

Scalable Bus Management

- Student ridership on HDPT buses **declines sharply** after **7 p.m.**, leading to **underutilized** service during evening hours
- An **excessive number** of bus stops increases fuel consumption and extends route times, **reducing overall system efficiency**

Solution

- **Use Smaller Vehicles**: During off-peak times, have JMU start to use smaller buses or vans
- **Consolidating Bus Stops Strategically:** Using the JMU Bus app, integrate a **GPS tracker**. This will track the most & least used bus stops to understand student usage of the buses

Impact

- Smaller vehicles are **more fuel-efficient** and require **less maintenance**, significantly **cutting fuel expenses** and **reducing long-term upkeep costs**
- GPS data enables **strategic stop consolidation**, which speeds up service, reduces total trip time, and improves bus utilization

Supporting Data

5>	A transportation study noted that **after 7pm** there are **fewer than 5 people** that ride the bus at a time.
30%	Smaller buses on streamlined routes after 7 p.m. can optimize JMU transit and **reduce operating costs** by **20–30%**.
48%	Implementing smaller buses during off-peak hours can cut transit costs by **32–57%, averaging 48% in savings.**

Looking Forward

JMU can explore applying for **EPA grants**, which offer free electric or hybrid buses. This can potentially save the university **$120,000–$150,000** over time and approximately **$6,400 annually** in operating costs.

Operational Strategy

Key Operational Inefficiencies

Imbalanced Route Utilization	**Overcrowding** on high-traffic routes while other routes remain **underutilized**, leading to **inefficient resource allocation**
Inconsistent Scheduling & Wait Times	Lack of alignment between bus **frequency** and **peak demand** results in long wait times for riders and **decreased reliability** of the system

Supporting Data

25%	15-30%	41%	51%
Dynamic scheduling can reduce wait times by up to 25%	App-based services can increase overall transit efficiency by 15-30%	41% of survey respondents are dissatisfied with bus crowding	51% are dissatisfied with schedule, routes, or frequency

Solutions

Dynamic Scheduling

Use real-time data and predictive analytics to **reallocate buses** in response to peak demand, **reducing wait times** on high-traffic routes and **minimizing idle buses** on underutilized routes.

App Based Microtransit

Launch an on-demand microtransit service for off-campus students, using AI to group riders and **operate during off-peak times**. This utilizes existing minivans/cutaway buses and has been successfully tested at colleges like **Georgia Tech** and **Texas A&M.**

Impact

The proposed changes will optimize bus utilization, reduce wait times, and improve service efficiency, leading to **increased service reliability**, **greater rider satisfaction**, and **enhanced accessibility** for off-campus students.

Operational Strategy (Implementation)

Dynamic Scheduling

Data Integration
Connect HDPT's GPS, and **route data** into a scheduling platform

AI Setup
Deploy an **AI-based scheduling tool** (e.g., Remix or Swiftly)

Training
Train staff to act on AI recommendations

Pilot Launch
Pilot on **high-volume routes** (e.g., ICS) to evaluate effectiveness

App Based Microtransit

App Design
Partner with **SMAD, CS, and ISAT students** to co-develop app

API Integration
Connect student-designed frontend with APIs from tools like **TransLoc**

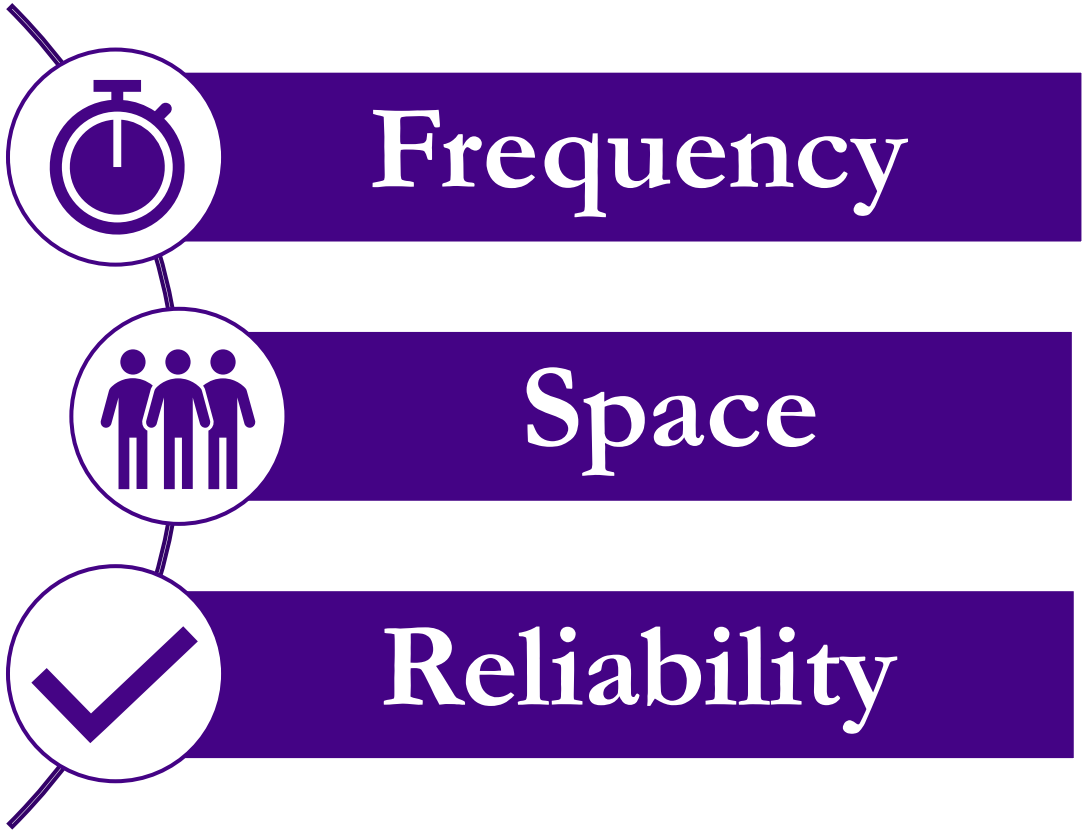
Fleet
Pilot with **Existing** JMU Cutaway Buses or Vans

Pilot & Promote
Pilot rollout + marketing via JMU digital channels

Measurability

Key Performance Indicators

Load Factor	Identify **overcrowded** routes and **underutilized** routes that can be optimized, ensuring a **more cost-effective and reliable** system.
On-Time Performance	**Identify delays, optimize scheduling, and improve service efficiency** to enhance rider satisfaction and operational effectiveness.
Cost per Rider	Measuring the **cost of service per passenger**, allowing us to identify cost-saving opportunities **while maintaining reliability and service quality.**
Customer Satisfactory Rating	Measure **rider experience and service perception**, enabling us to identify weak points, enhance transit operations, and improve overall user satisfaction.



Closing Thoughts

Strategic Direction

- Aligns improvements with **growing rider demand**
- Implements a **flexible approach** to meet the future needs of both the HDPT and JMU riders

Overview & Impact

- Projected minimum increase of **15% in service reliability** and a **48% boost in rider satisfaction**
- **Reduced wait times** and improved accessibility through Dynamic Scheduling and App-based micro-transit
- **Enhances customer satisfaction** while **maintaining financial efficiency**

Wrap Up

- This strategy makes a substantial impact on the HDPT with a focus on **efficiency** and **reliability**
- Other top priorities include **frequency** and **space** which ensures **rider satisfaction**
- This approach, overall, provides a strong foundation backed by **growth** and **adaptability**

Questions?







Sources:

- *Electric School Buses: Why Schools Should Transition to EV Fleets*. Electrum Charging, 2024. https://connect.electrumcharging.com/blog/electric-school-buses-why-schools-should-transition-to-ev-fleets.
- *JMU Student Survey - HDPT Bus System*. Google Forms, created by Grady Allen, 2025. https://forms.gle/Y3oink1cCk18W3NY8.
- *Public Transportation: Market Data & Analysis*. Statista Market Insights, 2024. https://www.statista.com/outlook/mmo/shared-mobility/public-transportation.
- *Transit Strategic Plan: Updated 2024*. Harrisonburg Department of Public Transportation, 2024. https://www.harrisonburgva.gov/sites/default/files/HDPT/files/TSP-Updated-2024.pdf.